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Business Internet Connection Types Cheat Sheet

Fibre Optic

What it is: High-speed internet delivered through fiber optic cables using light signals.

Speeds: Up to 10+ Gbps download and upload (commonly 100 Mbps - 1 Gbps for business)

How it works: Data transmitted as light pulses through glass or plastic fibers

Best for:

- Large businesses with high bandwidth needs
- Video conferencing and streaming
- Cloud-based applications
- Multiple users with heavy internet usage

Pros: Fastest speeds, reliable, symmetrical upload/download, low latency **Cons**: Most expensive, limited availability in rural areas

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ADSL (Asymmetric Digital Subscriber Line)

What it is: Internet delivered through existing copper telephone lines.

Speeds: Up to 24 Mbps download, up to 3 Mbps upload

How it works: Uses different frequencies on phone lines for voice and data

Best for:

- Small businesses with basic internet needs
- Light web browsing and email
- Areas where other options aren't available

Pros: Widely available, uses existing phone infrastructure, relatively affordable **Cons**: Slower speeds, speed decreases with distance from exchange, asymmetrical

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VDSL (Very-high-bit-rate Digital Subscriber Line)

What it is: Faster version of DSL using copper lines with fiber backhaul.

Speeds: Up to 100 Mbps download, up to 40 Mbps upload

How it works: Combines fiber optic cables to neighborhood nodes with copper to premises

Best for:

- Medium businesses needing faster than ADSL
- Video streaming and cloud applications
- Multiple users with moderate usage

Pros: Faster than ADSL, more widely available than pure fiber, better upload speeds **Cons**: Still distance-dependent, not as fast as pure fiber

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Cable/Coaxial

What it is: Internet delivered through the same coaxial cables used for cable TV.

Speeds: Up to 1000+ Mbps download, up to 50 Mbps upload

How it works: Uses different frequencies on coaxial cables for TV and internet

Best for:

- Businesses in urban/suburban areas
- High download requirements
- Streaming and large file downloads

Pros: High download speeds, widely available in populated areas **Cons**: Asymmetrical speeds, shared bandwidth with neighbors, can slow during peak times

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Satellite Internet

What it is: Internet delivered via communication satellites in space.

Speeds: Up to 100+ Mbps download, up to 20 Mbps upload

How it works: Data transmitted between satellite dish, satellite, and ground stations

Best for:

- Rural businesses with no other options
- Remote locations
- Backup internet connection

Pros: Available virtually anywhere, good for remote areas **Cons**: High latency, weather-dependent, data caps, expensive

Wireless/Fixed Wireless

What it is: Internet delivered through radio waves from towers to receiver antennas.

Speeds: Up to 1000 Mbps download and upload (varies by provider)

How it works: Radio signals transmitted between towers and customer premises equipment

Best for:

- Businesses in areas with good tower coverage
- Quick deployment needs
- Areas where cable installation is difficult

Pros: Fast deployment, no cables needed, can offer good speeds **Cons**: Line-of-sight requirements, weather-sensitive, coverage dependent

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Dedicated/Leased Lines

What it is: Private, dedicated connection between business and internet provider.

Speeds: Customizable from 1 Mbps to 10+ Gbps

How it works: Exclusive connection not shared with other customers

Best for:

- Large enterprises
- Businesses requiring guaranteed bandwidth
- Mission-critical applications
- High-security requirements

Pros: Guaranteed speeds, symmetrical, high reliability, dedicated support **Cons**: Very expensive, longer installation time, overkill for small businesses

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Quick Comparison

Type	Max Speed	Upload Speed	Availability	Cost	Best For
Fibre	10+ Gbps	Symmetrical	Urban/Suburban	High	Large business
ADSL	24 Mbps	3 Mbps	Very High	Low	Small business
VDSL	100 Mbps	40 Mbps	High	Medium	Medium business
Cable	1000+ Mbps	50 Mbps	Urban/Suburban	Medium	High download needs
Satellite	100+ Mbps	20 Mbps	Universal	High	Remote locations
Wireless	1000 Mbps	Symmetrical	Coverage areas	Medium	Quick deployment
Dedicated	10+ Gbps	Symmetrical	On request	Very High	Enterprise

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Choosing the Right Connection

Consider these factors:

- **Number of users**: More users need higher bandwidth
- **Applications**: Cloud apps, video calls, and file transfers need more speed
- **Upload requirements**: Video conferencing and cloud backups need good upload speeds
- **Reliability needs**: Mission-critical businesses need dedicated or fiber connections
- **Location**: Rural areas may be limited to satellite or wireless options
- **Budget**: Balance cost with performance requirements